

Resource Allocation System

Iteration 29 – Multi-Room Exam Best-Fit Strategy

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1 Iteration Goal

The goal of Iteration 29 is to improve multi-room exam allocation by adding a best-fit strategy. Iteration 28 introduced `MultiRoomExamGreedyStrategy`, which allowed splittable exam requests to use multiple classrooms. However, that strategy selected classrooms in loaded order, which could waste seating capacity.

Iteration 29 adds `MultiRoomExamBestFitStrategy`. This strategy still processes normal requests using existing greedy behavior, but for splittable `ExamRequest` objects it selects a room combination that reduces unused capacity.

This is not a global optimization strategy. It is a best-fit heuristic for one splittable exam request at a time.

2 Changes from Iteration 28

Iteration 28 introduced the first room-pool sharing strategy for exams. Iteration 29 keeps that behavior but improves room selection for splittable exams.

The main changes are:

- Added `MultiRoomExamBestFitStrategy.h`.
- Added `MultiRoomExamBestFitStrategy.cpp`.
- Updated `AllocationStrategyFactory` to support `multi_room_exam_best_fit`.
- Updated `DataController` config validation to accept `multi_room_exam_best_fit`.
- Updated `data/config.txt` so the strategy can be selected externally.
- Added a classroom example in `data/spaces.csv` for best-fit testing.
- Added a splittable exam request in `data/requests.csv` for best-fit testing.

3 Scenario

A splittable exam request may need more seats than one classroom can provide. The greedy multi-room strategy can allocate several rooms, but it may choose rooms in a way

that leaves unnecessary unused capacity.

For example, Request 26 needs 60 seats.

Strategy	Selected Rooms	Total Capacity	Waste
Multi-room greedy	B201 + B202	75	15 unused seats
Multi-room best-fit	B201 + B203	60	0 unused seats

This shows that best-fit can reduce wasted classroom capacity while still assigning enough seats for the exam.

4 Modules and Their Tasks

Module	Task in Iteration 29
<code>MultiRoomExamBestFitStrategy</code>	Implements the new best-fit room selection algorithm for splittable exam requests.
<code>GreedyAllocationStrategy</code>	Continues to provide normal processing behavior for non-exam requests and non-splittable exams.
<code>AllocationStrategyFactory</code>	Creates or returns <code>MultiRoomExamBestFitStrategy</code> when the config value is <code>multi_room_exam_best_fit</code> .
<code>DataController</code>	Accepts <code>multi_room_exam_best_fit</code> as a valid allocation strategy name from <code>config.txt</code> .
<code>ExamRequest</code>	Provides the exam time slot, participant count, and <code>canSplitAcrossRooms</code> flag used by the strategy.
<code>Allocation</code>	Stores the room-level <code>assignedParticipants</code> value for each split exam allocation.
<code>spaces.csv</code> and <code>requests.csv</code>	Provide a test case where best-fit produces less wasted capacity than loaded-order greedy selection.

5 Multi-Room Exam Best-Fit Design

The new strategy follows the same high-level structure as the previous multi-room strategy. Normal one-time and recurring requests are delegated to existing greedy behavior. Non-splittable exam requests are also processed like one-time exam requests.

For splittable exam requests, `requestedSpace` is not treated as the only allowed room. It remains in the CSV for compatibility and as a reference or preferred space. The strategy chooses suitable classrooms from the full loaded space pool.

Candidate classrooms must satisfy these constraints:

- The space type is **Classroom**.
- The room is available and not under maintenance.
- The required building matches when a building is specified.
- The required feature matches when a feature is specified.
- The requester is allowed to use classroom spaces.
- The room has no overlapping allocation at the exam time.

After candidate rooms are found, the strategy selects a combination whose total capacity is at least the exam participant count. The preferred combination is selected using these rules:

- Minimize unused capacity first.
- If there is a tie, prefer fewer rooms.
- If there is still a tie, keep deterministic behavior based on candidate order.

Allocations are not committed immediately. The strategy first builds the selected room combination. If enough capacity exists, the exam is approved and all allocations are created. If not enough capacity exists, the exam is rejected and no partial allocation rows are created.

Same-room sharing is still not allowed. Two overlapping exams cannot use the same physical room at the same time.

6 Iteration Comparison

Iter.	Focus	Added / Modified	Design bility	Responsi-	Prototype Effect
25	Exam requests	ExamRequest	Represented exams as a first-class request type.		Exam-specific data could be loaded, printed, and exported.
26	Request-type permission	RequestTypeRule	Separated request-type authorization from space-type authorization.		Students and staff could be rejected from submitting exam requests.
27	Allocation preparation	assignedParticipants and space pool access	Prepared allocations and strategy interfaces for multi-room exam allocation.		Strategies gained access to all spaces and allocations stored room-level participant counts.
28	Multi-room greedy	MultiRoomExamGreedyStrategy	Allowed splittable exams to use several classrooms from the room pool.		Large exams could be approved across multiple classrooms.
29	Multi-room best-fit	MultiRoomExamBestFitStrategy	Implemented room combination selection for splittable exams.		The system can reduce wasted classroom capacity compared with loaded-order greedy selection.

7 Updated Class Diagram

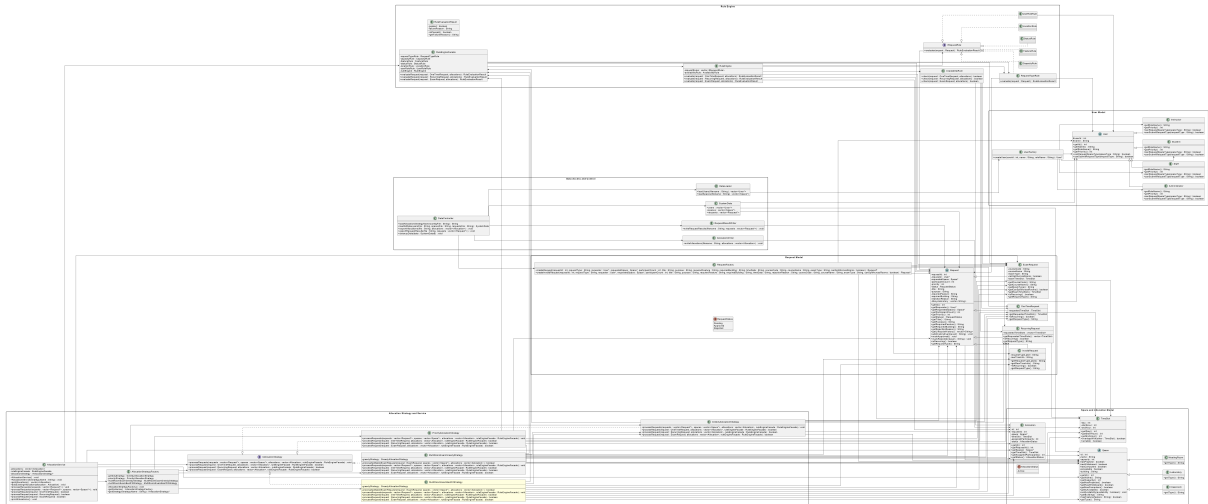


Figure 1: Updated class diagram after adding `MultiRoomExamBestFitStrategy`.

8 Updated Workflow Diagram

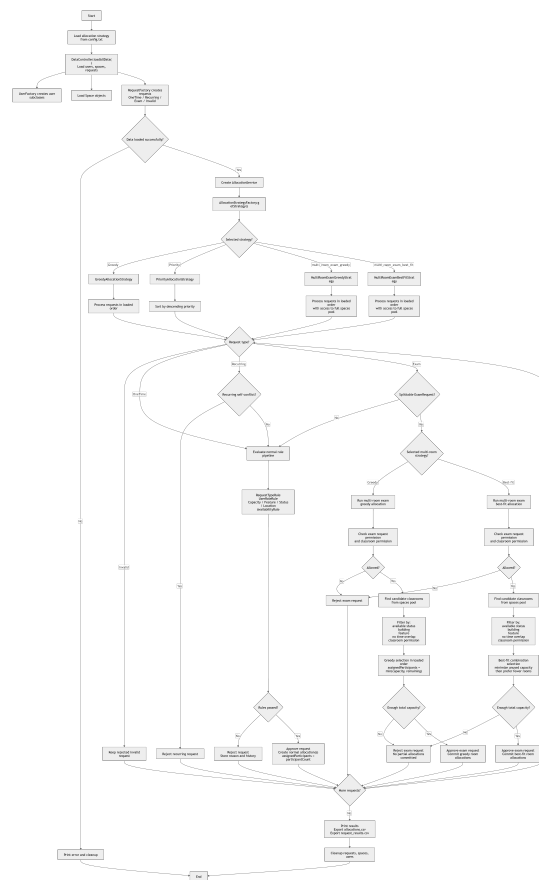


Figure 2: Updated workflow for multi-room exam best-fit allocation.

9 Prototype Implementation

The new strategy can be selected through the external config file:

```
allocation_strategy=multi_room_exam_best_fit
```

When selected, `AllocationService` receives the strategy from `AllocationStrategyFactory`. Normal requests continue through greedy behavior. Splittable exam requests use the full classroom pool and choose a best-fit room combination.

The prototype data includes Request 26, which needs 60 seats. The multi-room greedy strategy selects B201 and B202, giving 75 total seats and 15 unused seats. The best-fit strategy selects B201 and B203, giving exactly 60 seats and no unused seats.

The generated `allocations.csv` output records one allocation row per selected room. Each row includes `assignedParticipants`, showing how many students were assigned to that room.

10 Summary

Iteration 29 improves the exam allocation model by adding `MultiRoomExamBestFitStrategy`. The new strategy uses the Strategy Pattern to introduce a better room selection algorithm without rewriting `AllocationService` or the existing allocation strategies.

This iteration improves splittable exam allocation by reducing unused classroom capacity. It does not implement priority-best-fit, same-room sharing, request rescheduling, or global optimization. Existing greedy, priority, and multi-room greedy strategies remain available, and rule-based authorization continues to work through `RequestTypeRule` and `UserRoleRule`.